



Dresden Nuclear Power Station
6500 North Dresden Road
Morris, IL 60450

SVPLTR # 19-0019

10 CFR 50.73

March 29, 2019

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Dresden Nuclear Power Station, Unit 2 and 3
Renewed Facility Operating License Nos. DPR-19
NRC Docket No. 50-237 and 50-249

Subject: Licensee Event Report 237/2019-001-00, Secondary Containment Differential Pressure Exceeded Technical Specification due to Excess Ventilation Supply Flow

Enclosed is Licensee Event Report 237/2019-001-00, "Secondary Containment Differential Pressure Exceeded Technical Specification due to Excess Ventilation Supply Flow." This report describes an event which is being reported in accordance with 10 CFR 50.73(a)(2)(v)(C), "Any event or condition that could have prevented the fulfillment of the safety function of ... systems that are needed control the release of radioactive material."

There are no regulatory commitments contained in this submittal.

Should you have any questions concerning this letter, please contact Mr. Bruce Franzen at (815) 416-2800.

Respectfully,

A handwritten signature in dark ink, appearing to read "Joe Quinn".

Joe Quinn for
Peter J Karaba
Site Vice President
Dresden Nuclear Power Station

Enclosure Licensee Event Report 237/2019-001-00

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – Dresden Nuclear Power Station

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NRR

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollect.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Dresden Nuclear Power Station, Unit 2

2. Docket Number

05000237

3. Page

1 OF 3

4. Title

Secondary Containment Differential Pressure Exceeded Technical Specification due to Excess Ventilation Supply Flow

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
01	30	2019	2019	001-	00	03	29	2019	Dresden Unit 3	05000249
									Facility Name	Docket Number
									N/A	05000

9. Operating Mode

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)		

12. Licensee Contact for this LER**Licensee Contact**

Bruce Franzen, Regulatory Assurance Manager

Telephone Number (Include Area Code)

815-416-2800

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
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14. Supplemental Report Expected☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No**15. Expected Submission Date**

Month Day Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On January 30, 2019, the operating heating boiler tripped during a period of extremely cold external air temperatures. To address the potential of freezing the heating steam coils and filters, procedures provided direction to open the ventilation inlet plenum doors. At 0910, the Reactor Building differential pressure degraded and was determined to be below 0.25 inches water column vacuum. Technical Specification (TS) 3.6.4.1, "Secondary Containment," was not met and Condition A, Secondary Containment inoperable, was required since differential pressure degraded below the 0.25 inches water column vacuum by local average gauge readings. Standby Gas Treatment was started to assist in differential pressure control. At 0928, differential pressure improved and was maintaining above 0.25 inches water column vacuum. TS 3.6.4.1 Condition A was exited. This event is reportable under 10 CFR 50.73(a)(2)(v)(C), "Any event or condition that could have prevented the fulfillment of the safety function of ... systems that are needed control the release of radioactive material." The cause of the event was determined to be inadequate procedural guidance for responding to loss of heating boilers during extreme cold external air conditions. Corrective actions include revising station procedure to prioritize actions to mitigate ventilation transients during a loss of heater boilers.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Dresden Nuclear Power Station, Unit 2	05000237	YEAR	SEQUENTIAL NUMBER	REV NO.
		2019	- 001	- 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

Dresden Nuclear Power Station (DNPS), Unit 2, is a General Electric Company Boiling Water Reactor with a licensed maximum power level of 2957 megawatts thermal. The Energy Industry Identification System codes used in the text are identified as [XX].

A. Plant Conditions Prior to Event:

Unit: 02	Event Date: 01/30/2019	Event Time: 0910 CST
Reactor Mode: 1	Mode Name: Power Operation	Power Level: 100 percent

B. Description of Event:

On January 30, 2019, the operating heating boiler [BLR] tripped during a period of extremely cold external air temperatures of approximately -25 degrees Fahrenheit (F). The heating boiler tripped with subsequent supply low air temperature alarm [TA] at 0850. The heating boiler tripped due to sluggish operation of its pressure regulator [RG]. Due to the loss of heating steam to the ventilation supply steam coils, the temperature of the supply air to the Reactor Building (RB) [NG] ventilation [VA] system rapidly decreased, which began to degrade the ability of RB ventilation to maintain Secondary Containment [NG] differential pressure (DP). To address the potential of freezing the heating steam coils [HCL] and filters [FLT], station procedures for loss of heating boilers provided direction to open the RB ventilation inlet plenum doors [DR]. Opening the plenum doors further degraded the ability to maintain RB DP due to additional air supply path to the plenum vent fan suctions.

At 0900, there was an unexpected low RB differential pressure alarm [PDA]. At 0910, DP readings from the local gauges located on each wall of the refuel floor were obtained. The average of these RB DP readings was determined to be below 0.25 inches water column (wc) vacuum showing Secondary Containment DP had degraded. This condition led to not meeting Technical Specification (TS) 3.6.4.1, "Secondary Containment," and required entering Condition A, "Secondary Containment inoperable in Mode 1, 2, or 3," Required Action A.1, "Restore Secondary Containment to OPERABLE status," with a Completion Time of four hours.

At 0922, the Standby Gas Treatment (SBGT) [BH] system was started. At 0928, Unit 3 RB ventilation system was secured. RB DP improved and was maintaining above 0.25 inches wc vacuum. TS 3.6.4.1 Condition A was exited.

An 8-hour non-emergency notification was made via Emergency Notification System (ENS #53851) in accordance with 10 CFR 50.72(b)(3)(v)(C) for the loss of Secondary Containment.

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CONTINUATION SHEET**

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Dresden Nuclear Power Station, Unit 2	05000237	YEAR	SEQUENTIAL NUMBER	REV NO.
		2019	- 001	- 00

NARRATIVE**C. Cause of Event:**

The cause of the event was determined to be inadequate procedural guidance for responding to loss of heating boilers during extreme cold external air temperatures.

D. Safety Analysis:

Except for the heating boilers pressure regulator prior to the event, all equipment responded as designed. Operator action to start SBGT and to secure Unit 3 RB ventilation in accordance with procedure guidance prevented further degradation of the RB DP. Based on the Secondary Containment DP remaining negative during this event, the safety function of Secondary Containment was maintained, and this event is of low safety significance.

This event is being reported under 10 CFR 50.73(a)(2)(v)(C), "Any event or condition that could have prevented the fulfillment of the safety function of ... systems that are needed control the release of radioactive material."

E. Corrective Actions:

The operational response procedure for the loss of plant heating steam during extreme external air temperature conditions will be revised to prioritize Main Control Room operations to mitigate ventilation transients during a loss of heating boilers.

F. Previous Occurrences:

A previous event, Dresden LER 237/2016-001-00 on docket 50-237, was found during the investigation where a Secondary Containment DP transient followed a heating boiler shutdown. The Secondary Containment DP transient reported under this LER 2019-001-00 did not have the same causal factors.

G. Component Failure Data:

Not applicable.